Fiscal Year:	FY 2023	**Task Last Updated:**	FY 09/11/2025
PI Name:	McGregor, Heather Ph.D.		
Project Title:	Investigating Plantar Somatosensory Noise as a Countermeasure for Balance and Locomotion Impairments in Simulated Lunar and Martian Gravity		
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Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 TRISH-RFA-2101-PD: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Seidler, Rachael Ph.D. (MENTOR: University of Florida, Gainesville)		
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Task Description:	POSTDOCTORAL FELLOWSHIP The vestibular system in our inner ear is key to our sense of orientation and balance. However, in space, the vestibular system does not function normally and the brain cannot interpret its signals. The brain gradually reinterprets vestibular signals to adapt to microgravity, but when astronauts re-encounter gravity post-flight, they have difficulty maintaining their balance while standing and walking. This is because the brain needs time to readjust to normal vestibular signals in gravity. Astronauts require days or even weeks to regain their full sense of balance in gravity. This is problematic for missions to the Moon and Mars as poor balance may impair spacecraft evacuation or result in falls during extravehicular activities. While their vestibular system is dysfunctional, astronauts can use other sources of sensory information for maintaining balance in gravity. Indeed, research suggests that astronauts rely on somatosensory cues – derived from their sense of touch and body position – for maintaining stable upright stance following spaceflight. For example, the brain can use somatosensory information such as the angle of the ankle joint and the sensation of pressure between the foot sole and the ground to help maintain balance. If astronauts rely on somatosensory signals from the feet for maintaining balance after spaceflight, could enhancing those somatosensory signals improve their balance? The primary goal of my proposed study is to address this question. In this ground-based study, I will use a body weight support system to mimic gravity on the Moon and Mars. Participants will perform balance and walking tests in Earth's gravity, in simulated Moon gravity, and in simulated Mars gravity. During a subset of the tests, I will enhance somatosensory inputs from the feet using vibrating shoe insoles. I will test if insole vibration can improve stability while standing and walking in simulated partial gravity. The secondary goal of my proposed study is to characterize how reduced gravity and insole vibration change how the brain processes sensory inputs from the feet. I will measure the brain's response to foot sensory inputs using electroencephalography (EEG). The brain's response to sensory inputs will be measured in Earth gravity, simulated Moon gravity, and simulated Mars gravity, both with and without insole vibration. I will test if reductions in gravity alter how the brain processes sensory inputs from the feet. Moreover, I will assess if insole vibration augments the brain's response to sensory inputs from the feet. If insole vibration indeed enhances balance, locomotion, and the brain's sensory processing in simulated Moon and Martian gravity, this method of sensory enhancement will warrant further investigation. In addition to contributing to our fundamental understanding of the contribution of somatosensory inputs to human balance and locomotion, the findings of this research will test if insole vibration could be an effective countermeasure for post-flight balance and locomotor impairments for future deep space exploration missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have successfully developed vibrating insoles. Behavioral testing will determine if delivering subsensory insole vibration can speed up adaptation of balance and locomotor control after gravitational transitions (i.e., Earth to simulated Martian gravity, and from simulated Martian gravity back to Earth gravity). Behavioral tests and analyses will examine whether and how subsensory insole vibration impacts sensory reweighting once adapted to the different gravity levels. If we find that subsensory insole vibration is indeed effective in speeding up adaptation of balance and locomotor control following gravity transitions, insole vibration will warrant further investigation as a potential countermeasure for balance and locomotion impairments during future space travel. If effective, vibrating insoles could help improve crewmember performance shortly after landing on the Moon or Mars. Since this is a subsensory, imperceptible countermeasure, it would not imposing additional cognitive demands on astronauts. Vibrating insoles would be a small, light-weight, simple and cost-effective means of providing sensory augmentation during deep space exploration missions. Insoles and tactors are easily customizable, reusable, and could be easily integrated into the boots of existing spacesuits. Behavioral tests and analyses of sensory reweighting and following adaptation to different gravity levels will contribute to fundamental knowledge for future studies developing somatosensory and proprioceptive countermeasures for post-flight balance and locomotion impairments.
Task Progress:	When astronauts re-encounter gravity following spaceflight, they exhibit declines in postural and locomotor control. Adaptation following gravitational transitions requires days to weeks. Behavioral evidence shows that astronauts increase their reliance on somatosensory inputs from their feet and ankles to maintain upright stance post-flight. Subthreshold stimulation to the foot sole has been shown to improve postural stability while standing on Earth. Here we test if augmenting foot somatosensory inputs using subthreshold insole vibration will accelerate adaptation of balance and locomotor control after gravitational transitions. We used a body weight support (BWS) system to simulate body loading changes that occur with gravitational transitions. Subjects first performed repeated balance and walking tasks in a baseline Earth gravity condition (1 g, 0% BWS) with no insole vibration applied. Next, subjects performed those same tasks in simulated Martian gravity (3/8 g, 62.5% BWS), allowing us to assess their time course of adaptation. Finally, subjects repeated the tasks in Earth gravity (1 g, 0% BWS) to assess deadaptation (pivot from Lunar gravity in proposal). Subjects in our experimental group received insole vibration as they adapted to Mars gravity and while deadapting in final Earth gravity condition. Subjects in our control group received no insole vibration. Subjects were instrumented with IMUs on their lumbar region and feet. Each gravity condition consisted of an identical series of repeated tests. Subjects performed 6 blocks of bipedal standing balance tests (3 20-s trials with the eyes open) and 1- min bouts of treadmill walking (pivot from overground walking in proposal). Interleaved with behavioral tasks, we assessed subjects' perceptual thresholds to electrical nerve stimulation applied at the ankles (pivot from SEPs in proposal). This allowed us to assess whether movement and/or sensitivity changed during adaptation. We assessed stability during standing balance and walking as the mean sway area per trial as measured by the IMU over the lumbar region; this is the area of an ellipse covering 95% of the acceleration of sway angles in the coronal and sagittal planes. We compared behavioral adaptation curves, perceptual changes, and sensory reweighting between the experimental and control groups to assess the efficacy of subthreshold insole vibration in aiding adaptation of balance and locomotion during gravitational transitions. Data analyses showed no group by condition effect in adaptation curves in simulated Martian gravity or upon deadaptation back to 1g. Similarly, there were no group by condition differences in sensory reweighting. We found no evidence that subthreshold insole vibration speeds up the adaptation of postural or locomotor control to simulated gravitational transitions.
Bibliography Type:	Description: (Last Updated: 09/11/2025)

Articles in Peer-reviewed Journals	McGregor HR, Hupfeld KE, Pasternak O, Beltran NE, De Dios YE, Bloomberg JJ, Wood SJ, Riascos RF, Reuter-Lorenz PA, Seidler RD. "Crewmember demographic factors and their association with brain and ocular changes following spaceflight." npj Microgravity. 2025 Aug 28;11:59. https://doi.org/10.1038/s41526-025-00505-9 ; PMID: 40877282 PMCID: PMC12394539 , Aug-2025